

User provisioning

Entering the identity provider market.

ROLE

Lead designer

TEAM

Product Manager · Engineers

TIMELINE

2025

STATUS

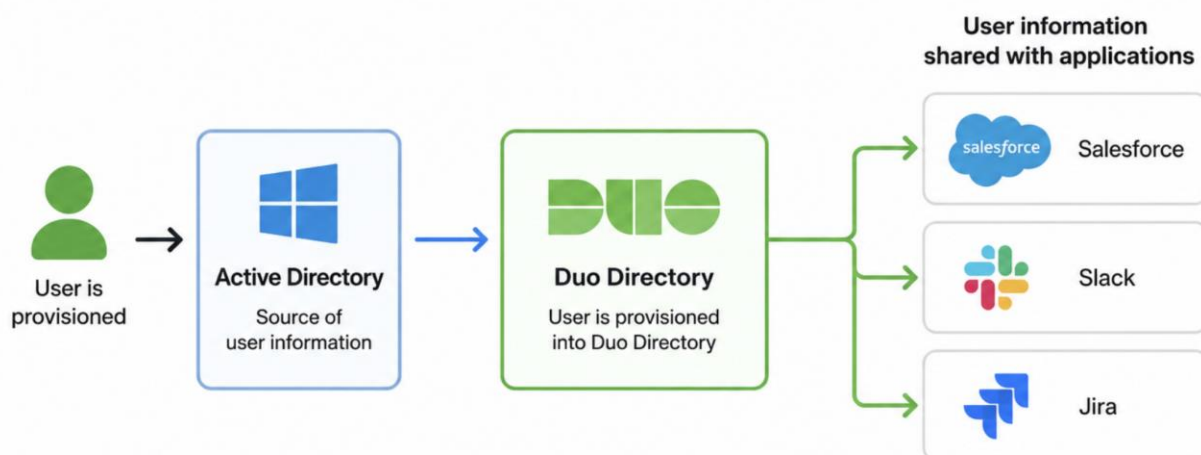
570+ customers · 181K+ users provisioned · ~1,000 active integrations

— OVERVIEW

Launch user provisioning in 6 months.

Provisioning, or outbound SCIM, is essential for identity management. It ensures user details like names, pronouns, and roles are automatically synced across applications. Our target users, IT admins, set up provisioning so their teams always have the right access.

Duo is a two-factor authentication system that verifies user identity. Expanding beyond security into the identity space meant building automated user provisioning from scratch. I was the lead designer for this section, tasked with delivering final designs within two months.



— PROCESS

Defining the minimum lovable product.

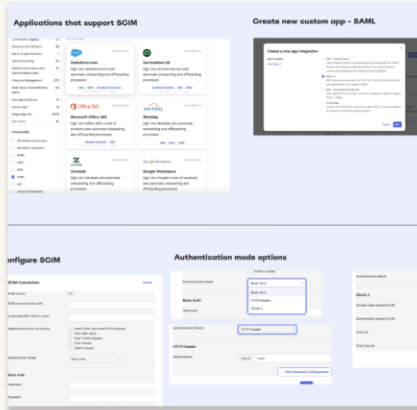
Finding the sweet spot between easier to build and most usable involved several targeted steps.

STEP 01

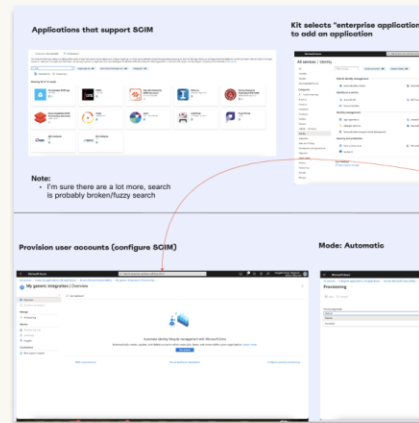
Studying multiple competitors

I set up accounts on Okta, Azure, and Jumpcloud to test user provisioning end-to-end, putting myself in the IT admin's shoes. By mapping each workflow I identified common steps and opportunities for improvement. Our PMs and engineers frequently referenced these maps and walk-through videos, and it was striking how differently each IdP modelled the same provisioning concept.

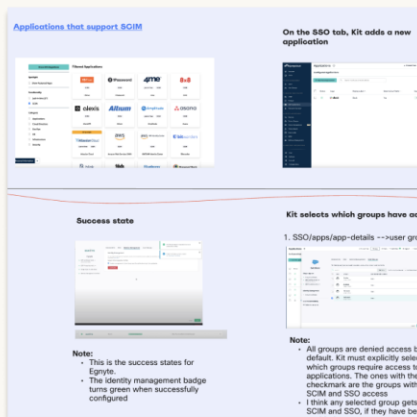
Okta



Azure



Jumpcloud



Competitive analysis · Okta · Azure · Jumpcloud workflow maps
STEP 02

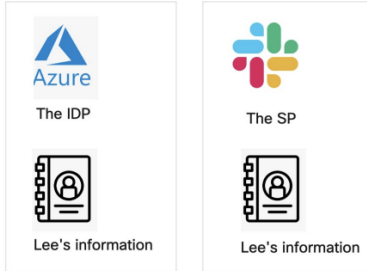
Creating a SCIM glossary for the team

Because SCIM was new to much of the team, I created a glossary explaining key provisioning concepts. It became a shared reference for engineering, product, and other design teams across the project, reducing misalignment and speeding up reviews.

Provisioning

A way to create, update and delete user accounts in other places/SPs

For Lee to be able to use resources, her information needs to be stored in two places:



Manual provisioning

- Kit logs into the SP and manually adds information about each Lee. He populates the SP directory
- Or
- Kit writes a custom plugin to pull in info from the IDP, where he sets the sync frequency
- Or
- Maybe this SP needs only email ID (provided while logging in). The SP asks Lee to fill out the rest of the information (e.g: Hey new person! could you please fill out your name and pronouns for us?)
- Or
- Import a CSV from the IDP, while configuring an app (Jumpcloud does this)

Automatic provisioning

Lee's details from the IDP are synced with the SP automatically. SCIM and JIT are some common protocols used for automatic provisioning

SCIM

System for Cross-domain Identity Management. A way to do automatic provisioning. This means any time Lee's information is changed in the IDP, it is immediately reflected in the SP.

SCIM is used to create Lee accounts at the SP. This means when Lee joins acme on day 1, an account for her is already created in the SP.

Although SAML/JIT can do something similar, where they create/update accounts, they cannot delete Lee. SCIM's main advantage is that users can be deprovisioned immediately

SSO/SAML

A way to authenticate into an application using a single username/password.

Summary: SSO and SCIM are best friends

What's the difference between SSO and SCIM?

SSO authenticates Lee. SCIM shares Lee's CRUD information

SSO handles - how to login, not Lee's account provisioning*

*well, actually it's the provision time that's different. With SSO/SAML, Lee's account *can* be created (not deleted/updated) during her first login to the SP. With SCIM, Lee's account is created as soon as she's registered in the IDP. And she can be deprovisioned automatically

Is SCIM another way to protect an application?

No. SCIM has nothing to do with authentication. All SCIM does is pass on Lee's information/attributes to the SP

Can you use SSO without user provisioning?

Generally, No. Lee cannot login without having an account.

But,

Let's say the SP just needs Lee's email ID to create an account. The SP has all the information required to create an account for Lee while she is logging in. This means Lee's account is created when she logs into the SP, not when she registered in the IDP. This is a rare use case.

Can you use SSO without SCIM?

Yes. But this is a rare use case.

Kit would have to either manually provision or use [JIT \(a SAML protocol\)](#), where the account is created. But changes to Lee's permissions/ Lee leaving will not be automatically reflected.

Can SCIM work without SSO?

Yes, but this means the IDP needs to save a username and a password version for EACH SP. 🤖

Or

Maybe there's a legacy app with just usernames, that Lee's info needs to be updated on this legacy app. Another very unlikely scenario

Does SCIM work with non-sso apps?

Yes. But this is another unlikely scenario

How are the user attributes being sent to the SP?

in Okta, once Kit has SCIM setup, they can go to the profile editor and edit attribute assignments

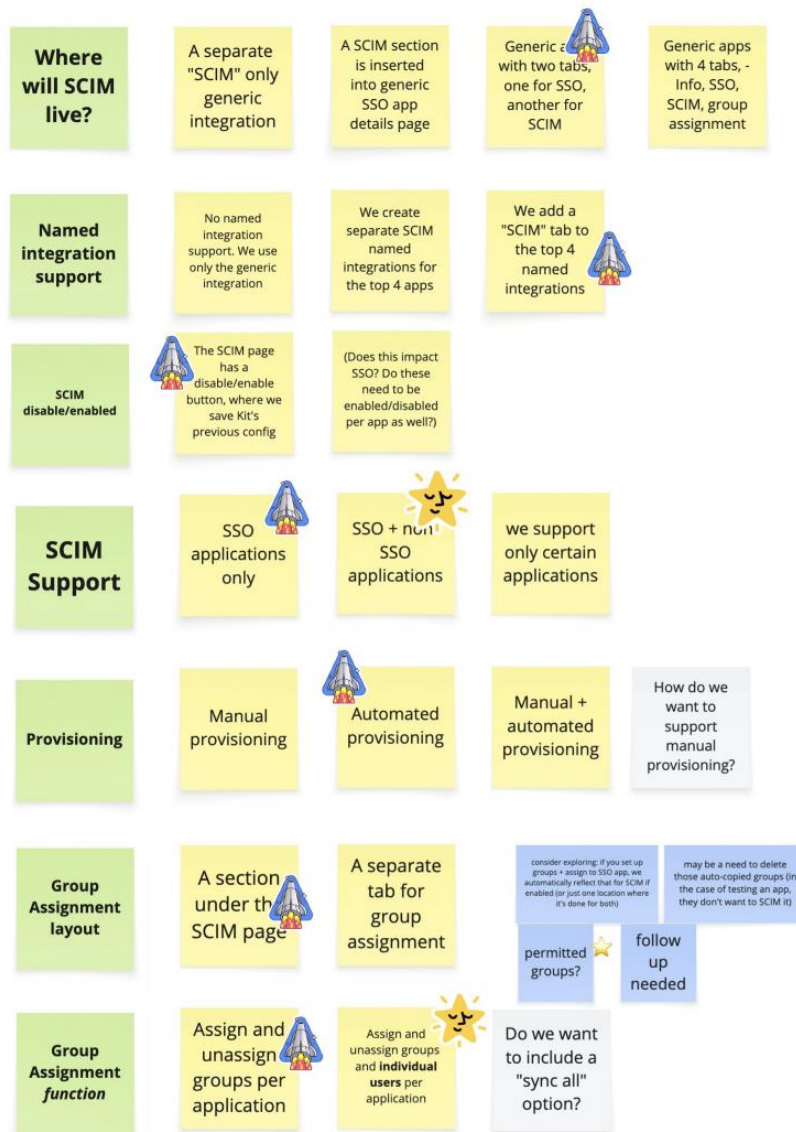


SCIM glossary · shared reference across engineering, product, and design
STEP 03

Mapping a feature-solution matrix

To answer "what's a feasible 6-month MVP?", I mapped every possible solution for each provisioning feature: where SCIM would live, named integration support, group assignment layout, and more. Together with PM and engineers we workshopped and voted on one scenario per feature, establishing our MVP scope.

Design Scoping



Scoping workshop · feature-solution matrix
STEP 04

Story-mapping to surface feasibility risks

Story mapping the three main structural options helped the team estimate technical effort. We discovered that touching legacy SSO pages would cost an extra six months. This narrowed us to an approach, a separate SCIM tab, that delivered the best experience without disturbing existing code.



Story mapping · three structural approaches and their technical costs
STEP 05

Challenging the assumption that SSO and provisioning groups are the same

We had assumed IT admins always assign identical user groups for SSO and provisioning, true in 90% of cases, but research uncovered real exceptions. Some admins named their SSO and provisioning groups differently. Others provisioned users into apps like office check-in systems where those users would never actually log in. The final design added separate group selectors and a "use existing SSO permitted groups" toggle to handle both cases.

Before research

Same groups for SSO and provisioning

Groups
Select existing groups that will receive updates from Duo in this application.

SSO permitted groups

Test1 × Testgroup × Admins × Managers × ▾

After research

Option to select provisioning specific groups

Groups
Select existing groups that will receive updates from Duo in this application.

 Users or groups will be automatically created, updated, and deactivated in this application.

Groups

Test1 × Testgroup × Admins × Managers × ▾

☐ **Use existing SSO permitted groups**
To change these groups, switch to the Single Sign-On tab > Permitted groups

☐ **Send group information to this application**
If unchecked, Duo will only send user information.

Group selector design · separate SSO and provisioning group assignment

STEP 06

Balancing consistency with different use-cases

Outbound provisioning was designed alongside an inbound SCIM project. The initial plan was to reuse the same attribute-mapping components across both, but I discovered that the mapping directions, and therefore the interaction logic, were fundamentally different. I worked with the inbound designer to add targeted variations to the shared pattern so both use-cases remained consistent without compromising either.

Key question: How similar do these experiences need to be?
How likely is it that the same Kit will configure both Inbound + Outbound SCIM?

ISCIM attribute mapping

1. SCIM attribute : Duo attribute is a 1:1 mapping

SCIM attribute	Duo user attribute	Action
username	User name	↔
email	Email ID	↔

2. Kit can create new Duo user attributes (and the source external directory is inbound SCIM/Okta connection)

En
Employee Grade
Employee Title
Employee
En + New attribute

3. Kit can only select from a list of predefined "SCIM" attributes.

En
employeeNumber
employeeNumber
En + New SCIM attribute

Provisioning attribute mapping

1. Kit needs to map one Duo attribute to multiple SCIM attributes. ✓

Duo user attribute	SCIM attribute	Action
Email ID	username	↔
Email ID	email	↔

2. Kit can't create new Duo user attributes (because Duo can't send a new attribute to the downstream, outgoing application) ✓

En
Employee Grade
Employee Title
Employee
En + New attribute

3. We cannot create new user attributes here, Kit will have to create and select user attributes

En
employeeNumber
employeeNumber
En + New SCIM attribute

Out of scope for MVP

Some SPs don't follow the standard SCIM spec (e.g. SCIM dev - username is required and different than the SCIM standard)

Required SCIM attributes generally fit the standard SCIM spec (even if we don't support everything)

Our definition of Generic SCIM = it fits the generic SCIM standard (not that it fits absolutely everything)

This also impacts/improves the SCIM developer experience.

Attribute mapping + Add mapping

Configure how Duo user attributes are mapped to SCIM attributes in your application so that user information is received in the correct format. To view or create Duo user attributes, go to [User Attributes](#).

Duo user attribute	SCIM attribute	Action
Email	username	↔
Email	email	↔
Given name	givenName	↔
Family name	familyName	↔
Username	Employee ID	↔

Cost Center
Department
Employee Grade
Employee Title
Employee

Attribute mapping + Add mapping

Configure how Duo user attributes are mapped to user attributes in Duo so that user information is received in the correct format. If you select a Duo user attribute that doesn't exist in your current schema, a new attribute can be created. To view all Duo user attributes, go to [User Attributes](#).

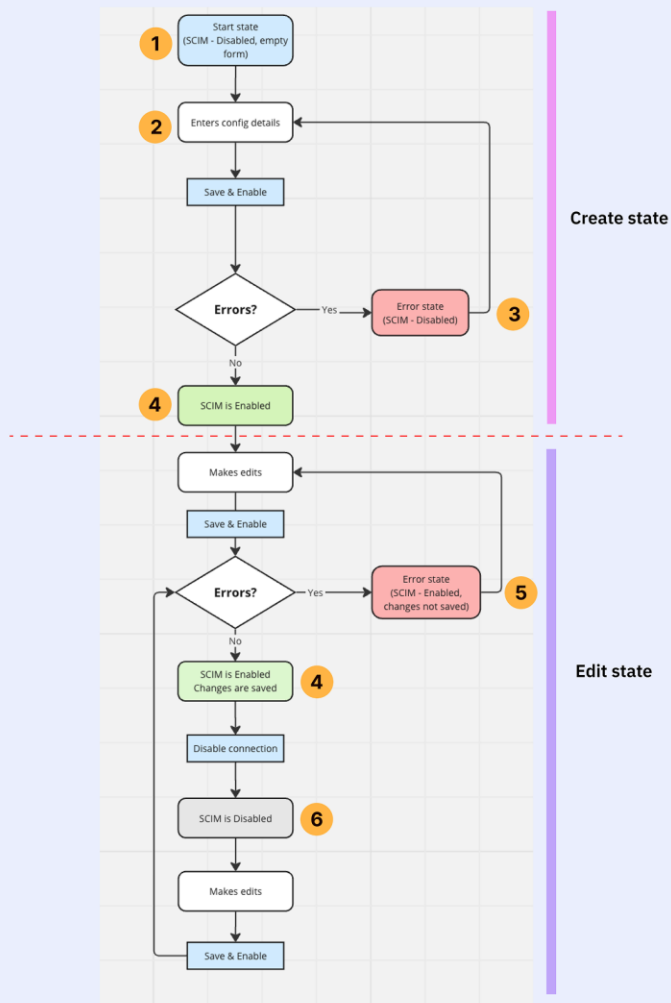
SCIM attribute	Duo user attribute	Action
username	Username	↔
email	Email	↔
displayName	Full Name	↔
department	Department	↔
employeeNumber	Employee Number (new attribute)	↔

Cost Center
Department
Employee Grade
Employee Title
Employee
Employee Number (new attribute) + Add

Attribute mapping · inbound vs outbound SCIM component variations
STEP 07

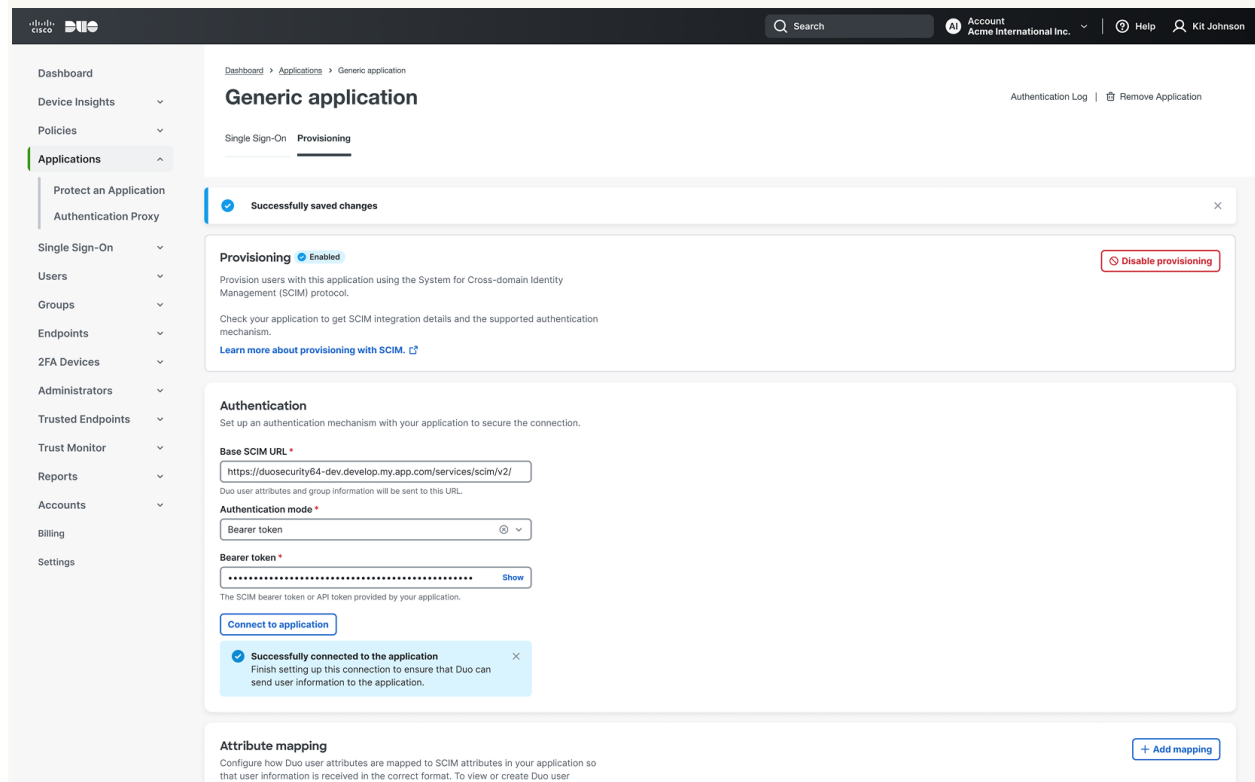
Differentiating create and edit states

To clearly define when components should behave differently in create versus edit state, I mapped out a flowchart covering the key decision points. This became a shared reference for the team, reducing ambiguity during handoff and keeping the interaction logic consistent across the provisioning flow.



Create vs edit state flowchart · shared reference for handoff
— FINAL DESIGNS

Final designs



Workday · Provisioning tab, shipped configuration UI
— IMPACT

Adopted at scale across the enterprise.

Outbound provisioning has been successfully adopted by 570+ customers, enabling large-scale automation of identity and access management across the ecosystem. This capability has supported the provisioning of 181K+ users and 3.3K+ groups through approximately ~1,000 active integrations.

Sangeetha has been the primary design stakeholder responsible for getting Outbound SCIM off the ground, which is one of the largest pillars of IDP functionality. From day 1, Sangeetha showed strong project leadership by coordinating with the Inbound SCIM team and finding an innovative way to build a Magnetized outbound SCIM product inside the non-Magnetized application details page (aka "Maglarro"). She has also helped our teams move conversations forward about future functionality like IGA by visualizing new flows we could build via SSO and Duo Central.

— Design Director

Sangeetha is a freakin rock star. I am legit blown away by how much she has put together on Outbound SCIM so quickly.

— Product Manager, Duo